

LAW DOME SUMMIT, Antarctica

WMO: 898110

Lat: **66.730S**Lon: **112.835E**Elev: **4488**StdP: **12.46**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	-40.6	-36.5	N/A	N/A	N/A	N/A	N/A	N/A	50.9	-9.9	46.1	-7.0	8.7	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	22.5	N/A	19.9	N/A	18.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 45.3	39.9	33.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.2	12.4	6.9	-1.4	-7.7	-12.4	-13.6	-13.2	-13.5	-12.9	-7.9	2.5	10.5		
	DBStd	12.45	4.27	6.11	7.06	9.05	10.08	8.90	10.26	9.39	9.19	7.29	6.38	4.52		
	HDD50	19798	1164	1206	1594	1730	1936	1909	1960	1968	1887	1795	1425	1224		
	HDD65	25273	1629	1626	2059	2180	2401	2359	2425	2433	2337	2260	1875	1689		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	11.8	15.0	13.4	13.1	12.2	7.0	8.7	6.7	6.1	12.5	15.5	15.6	15.9		
(15) Precipitation	PrecAvg	17.0	1.4	1.4	1.6	1.4	1.2	1.5	1.7	1.7	1.3	1.2	1.4	1.3		
	PrecMax	21.2	2.2	2.4	2.6	2.4	2.0	2.6	3.0	2.9	2.4	1.9	2.3	2.1		
	PrecMin	13.7	0.7	0.2	0.9	0.6	0.3	0.6	0.8	0.5	0.5	0.4	0.4	0.8		
	PrecStd	2.0	0.4	0.6	0.4	0.5	0.4	0.5	0.6	0.6	0.4	0.4	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.8	24.1	16.0	12.9	12.5	8.6	10.5	13.3	9.9	11.7	21.0	27.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	22.4	19.7	12.6	10.2	9.6	3.3	6.3	7.5	6.4	7.4	16.6	22.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	20.2	17.5	10.5	7.0	3.9	0.3	3.4	2.5	3.1	4.0	13.9	19.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	18.4	15.7	8.6	4.2	0.7	-2.5	0.2	-0.8	0.0	1.7	11.4	16.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range		MDBR	9.6	10.9	11.2	11.8	11.0	9.8	9.8	9.3	11.2	11.5	11.0	10.1		
	5% DB	MCDBR	12.6	13.3	12.8	13.0	10.9	13.9	10.0	10.4	13.1	13.8	14.8	15.6		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub	0.214	0.202	0.185	0.151	0.102	N/A	0.105	0.149	0.177	0.185	0.200	0.214	0.214		
	taud	2.237	2.260	2.270	2.163	1.904	N/A	1.896	2.134	2.246	2.268	2.250	2.220	2.220		
	Ebn at Noon	336	324	301	248	132	0	117	244	299	329	340	340	340		
	Edh at Noon	37	31	23	13	4	0	3	13	23	31	36	39	39		
(44) All-Sky Solar Radiation	RadAvg	2686	1899	995	338	51	0	19	187	705	1558	2489	2981	2981		
	RadStd	51	48	31	14	3	0	1	10	21	35	66	36	36		

Historical Trends

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